

NAME : P SAI VENKAT RAJU

SUBJECT : AI ASSISTED CODING

ROLL NO : 2403A510G0

ASSIGNMENT : 6.1

Task Description #1 (Classes – Employee Management)

- Task: Use AI to create an Employee class with attributes (name, id, salary) and a method to calculate yearly salary.
- Instructions:
 - Prompt AI to generate the Employee class.
 - Analyze the generated code for correctness and structure.
 - Ask AI to

Expected Output #1:

- A class with constructor, `display_details()`, and `calculate_bonus()` methods.

PROMPT :

To create an Employee class with attributes (name, id, salary) and a method to calculate yearly salary.

Add a method to give a bonus and recalculate salary.

CODE :

```
palindrome.py cart.py appinal.py aic.py x date.py nw.py doctor.py assignment1.1.file
aic.py > Employee
1 class Employee:
2     def __init__(self, emp_id, name, salary):
3         """
4         Initializes an Employee with an id, name, and salary.
5         :param emp_id: Unique identifier for the employee.
6         :param name: Name of the employee.
7         :param salary: Monthly salary of the employee.
8         """
9         self.emp_id = emp_id
10        self.name = name
11        self.salary = salary
12
13    def display_details(self):
14        """
15        Displays employee details.
16        """
17        print(f"Employee ID: {self.emp_id}, Name: {self.name}, Salary: {self.salary}")
18
19    def calculate_yearly_salary(self):
20        """
21        Calculates the yearly salary assuming the given salary is monthly.
22        """
23        return self.salary * 12
24
25    def calculate_bonus(self, bonus_percent):
26        """
27        Adds a bonus to the current monthly salary and updates it.
28        :param bonus_percent: Bonus percentage to be added to the salary.
29        :return: New salary after adding bonus.
30        """
31        bonus_amount = self.salary * (bonus_percent / 100)
32        self.salary += bonus_amount
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```

```

44 class Employee:
45
46     def calculate_yearly_salary(self):
47         """
48         Calculates the yearly salary assuming the given salary is monthly.
49         """
50         return self.salary * 12
51
52     def calculate_bonus(self, bonus_percent):
53         """
54         Adds a bonus to the current monthly salary and updates it.
55         :param bonus_percent: Bonus percentage to be added to the salary.
56         :return: New salary after adding bonus.
57         """
58         bonus_amount = self.salary * (bonus_percent / 100)
59         self.salary += bonus_amount
60         return self.salary
61
62 # Example usage:
63 if __name__ == "__main__":
64     emp = Employee(emp_id=1, name="Alice", salary=5000)
65     emp.display_details()
66     print("Yearly Salary:", emp.calculate_yearly_salary())
67
68     # Give a 10% bonus to the employee and update the salary.
69     new_salary = emp.calculate_bonus(10)
70     print("New Salary after 10% bonus:", new_salary)

```

```

pranav@PRANAVs-MacBook-Air AI ASSISTED CODING % cd /Users/Shared/AI/ ASSISTED\ CODING ; /usr/bin/env /usr/local/bin/python3 /Users/pranav/.
vscode/extensions/ms-python.debugpy-2025.10.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 62279 -- /Users/Shared/AI/ AS
SISTED\ CODING/aic.py
Employee ID: 1, Name: Alice, Salary: 5000
Yearly Salary: 60000
New Salary after 10% bonus: 5500.0
Employee ID: 1, Name: Alice, Salary: 5000
Yearly Salary: 60000
New Salary after 10% bonus: 5500.0
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %

```

OUTPUT :

```

SyntaxError: invalid syntax
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING % cd /Users/Shared/AI/ ASSISTED\ CODING ; /usr/bin/env /usr/local/bin/python3 /Users/pranav/.
vscode/extensions/ms-python.debugpy-2025.10.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 62279 -- /Users/Shared/AI/ AS
SISTED\ CODING/aic.py
Employee ID: 1, Name: Alice, Salary: 5000
Yearly Salary: 60000
New Salary after 10% bonus: 5500.0
Employee ID: 1, Name: Alice, Salary: 5000
Yearly Salary: 60000
New Salary after 10% bonus: 5500.0
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %

```

Task Description #2 (Loops – Automorphic Numbers in a Range)

- Task: Prompt AI to generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.
- Instructions:
 - o Get AI-generated code to list Automorphic numbers using a for loop.
 - o Analyze the correctness and efficiency of the generated logic.
 - o Ask AI to regenerate using a while loop and compare both implementations.

Expected Output #2:

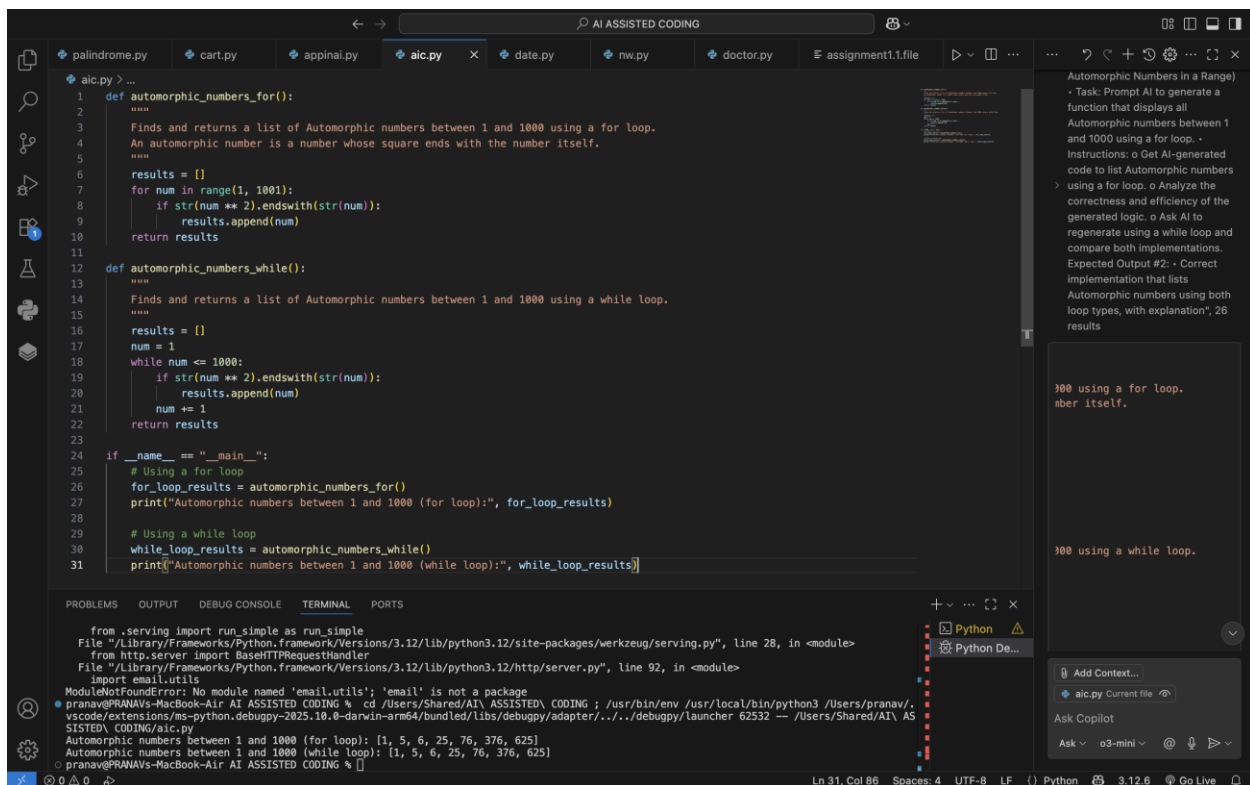
- Correct implementation that lists Automorphic numbers using both loop types, with explanation.

PROMPT :

To generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.

To regenerate using a while loop and compare both implementations.

CODE GENERATED :



```
1 def automorphic_numbers_for():
2     """
3     Finds and returns a list of Automorphic numbers between 1 and 1000 using a for loop.
4     An automorphic number is a number whose square ends with the number itself.
5     """
6     results = []
7     for num in range(1, 1001):
8         if str(num ** 2).endswith(str(num)):
9             results.append(num)
10    return results
11
12 def automorphic_numbers_while():
13    """
14    Finds and returns a list of Automorphic numbers between 1 and 1000 using a while loop.
15    """
16    results = []
17    num = 1
18    while num <= 1000:
19        if str(num ** 2).endswith(str(num)):
20            results.append(num)
21        num += 1
22    return results
23
24 if __name__ == "__main__":
25     # Using a for loop
26     for_loop_results = automorphic_numbers_for()
27     print("Automorphic numbers between 1 and 1000 (for loop):", for_loop_results)
28
29     # Using a while loop
30     while_loop_results = automorphic_numbers_while()
31     print("Automorphic numbers between 1 and 1000 (while loop):", while_loop_results]
```

Automorphic Numbers in a Range)

- Task: Prompt AI to generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.
- Instructions: o Get AI-generated code to list Automorphic numbers
- using a for loop. o Analyze the correctness and efficiency of the generated logic. o Ask AI to regenerate using a while loop and compare both implementations. Expected Output #2: - Correct implementation that lists Automorphic numbers using both loop types, with explanation", 26 results

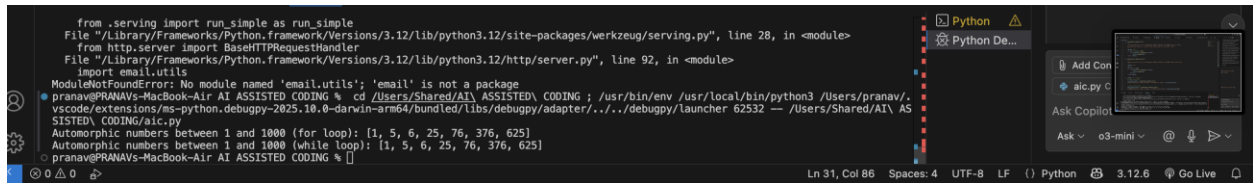
1000 using a for loop.
number itself.

1000 using a while loop.

from .serving import run_simple as run_simple
File "/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/werkzeug/serving.py", line 28, in <module>
from http.server import BaseHTTPRequestHandler
File "/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/http/server.py", line 92, in <module>
import email.utils
ModuleNotFoundError: No module named 'email.utils': 'email' is not a package
• pranav@PRANAVs-MacBook-Air AI ASSISTED CODING % cd /Users/Shared/AI/ ASSISTED CODING ; /usr/bin/env /usr/local/bin/python3 /Users/pranav/.vscode/extensions/ms-python.debugpy-2025.10.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 62532 -- /Users/Shared/AI/ ASSISTED CODING/aic.py
Automorphic numbers between 1 and 1000 (for loop): [1, 5, 6, 25, 76, 376, 625]
Automorphic numbers between 1 and 1000 (while loop): [1, 5, 6, 25, 76, 376, 625]
• pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %

Ln 31, Col 86 Spaces: 4 UTF-8 LF () Python 3.12.6 Go Live

OUTPUT :

A screenshot of a Visual Studio Code editor window. The main editor area shows a Python script with a syntax error. The error message is: "ModuleNotFoundError: No module named 'email.utils'; 'email' is not a package". The script is a simple HTTP server using Werkzeug. The Copilot sidebar on the right is open, showing a search for 'aic.py' and a snippet of code. The status bar at the bottom indicates the file is at line 31, column 86, with 4 spaces, UTF-8 encoding, and LF line endings. The Python version is 3.12.6, and the Go Live extension is active.

Task Description #3 (Conditional Statements – Online Shopping Feedback Classification)

- Task: Ask AI to write nested if-elif-else conditions to classify online shopping feedback as Positive, Neutral, or Negative based on a numerical rating (1–5).
- Instructions:
 - Generate initial code using nested if-elif-else.
 - Analyze correctness and readability.
 - Ask AI to rewrite using dictionary-based or match-case structure.

Expected Output #3:

- Feedback classification function with explanation and an alternative approach

PROMPT :

To write nested if-elif-else conditions to classify online shopping feedback as Positive, Neutral, or Negative based on a numerical rating (1–5)

CODE :

```
palindrome.py cart.py appinal.py aic.py x date.py nw.py doctor.py assignment1.1.file
def classify_feedback_if(rating):
    """
    Classifies online shopping feedback based on a numerical rating (1-5)
    using nested if-elif-else conditions.
    """
    Rating Criteria:
    - 4 or 5: Positive feedback.
    - 3: Neutral feedback.
    - 1 or 2: Negative feedback.
    - Any other value: Invalid rating.
    """
    if rating == 4 or rating == 5:
        return "Positive"
    elif rating == 3:
        return "Neutral"
    elif rating == 1 or rating == 2:
        return "Negative"
    else:
        return "Invalid rating"

def classify_feedback_match(rating):
    """
    Classifies online shopping feedback using the match-case structure
    available in Python 3.10+.
    """
    Rating Criteria:
    - 4 or 5: Positive feedback.
    - 3: Neutral feedback.
    - 1 or 2: Negative feedback.
    - Any other value: Invalid rating.
    """
    match rating:
        case 4 | 5:
            return "Positive"
        case 3:
            return "Neutral"
        case 1 | 2:
            return "Negative"
        case _:
            return "Invalid rating"

Testing dictionary mapping implementation:
Rating: 5 => Positive
Rating: 4 => Positive
Rating: 3 => Neutral
Rating: 2 => Negative
Rating: 1 => Negative
Rating: 0 => Invalid rating
Rating: 6 => Invalid rating
Rating: -1 => Invalid rating
Rating: 3 => Neutral
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %
```

o Generate initial code using nested if-elif-else.
o Analyze correctness and readability.
o Ask AI to rewrite using dictionary-based or match-case structure.
Expected Output #3:
• Feedback classification function with explanation and an alternative approach

Used 1 reference

numerical rating (1-5)

```
palindrome.py cart.py appinal.py aic.py x date.py nw.py doctor.py assignment1.1.file
def classify_feedback_match(rating):
    """
    Classifies online shopping feedback using the match-case structure
    available in Python 3.10+.
    """
    Rating Criteria:
    - 4 or 5: Positive feedback.
    - 3: Neutral feedback.
    - 1 or 2: Negative feedback.
    - Any other value: Invalid rating.
    """
    match rating:
        case 4 | 5:
            return "Positive"
        case 3:
            return "Neutral"
        case 1 | 2:
            return "Negative"
        case _:
            return "Invalid rating"

# Alternative approach using dictionary for direct mapping:
def classify_feedback_dict(rating):
    """
    Classifies online shopping feedback using a dictionary mapping.
    Note: This approach requires all valid ratings to be pre-defined.
    """
    mapping = {
        1: "Negative",
        2: "Negative",
        3: "Neutral",
        4: "Positive",
        5: "Positive"
    }
    return mapping.get(rating, "Invalid rating")

# Test cases for all implementations:
test_ratings = [5, 4, 3, 2, 1, 0, 6, -1, "3"]

print("Testing Nested if-elif-else implementation:")
for rating in test_ratings:
    try:
        # Convert rating to int if possible to handle numeric tests
        r = int(rating)
    except:
        r = rating

Testing dictionary mapping implementation:
Rating: 5 => Positive
Rating: 4 => Positive
Rating: 3 => Neutral
Rating: 2 => Negative
Rating: 1 => Negative
Rating: 0 => Invalid rating
Rating: 6 => Invalid rating
Rating: -1 => Invalid rating
Rating: 3 => Neutral
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %
```

o Generate initial code using nested if-elif-else.
o Analyze correctness and readability.
o Ask AI to rewrite using dictionary-based or match-case structure.
Expected Output #3:
• Feedback classification function with explanation and an alternative approach

Used 1 reference

numerical rating (1-5)

The screenshot shows a VS Code editor with a file named `aic.py`. The code defines a list of test ratings and tests three different classification functions: `classify_feedback_if`, `classify_feedback_match`, and `classify_feedback_dict`. The terminal output shows the results of these tests for ratings 5, 4, 3, 2, 1, 0, and -1.

```
58 # Test cases for all implementations:
59 test_ratings = [5, 4, 3, 2, 1, 0, -1, "3"]
60
61 print("Testing Nested if-elif-else implementation:")
62 for rating in test_ratings:
63     try:
64         # Convert rating to int if possible to handle numeric tests
65         r = int(rating)
66     except Exception:
67         r = rating
68     print(f"Rating: {rating} => {classify_feedback_if(r)}")
69
70 print("\nTesting match-case implementation:")
71 for rating in test_ratings:
72     try:
73         r = int(rating)
74     except Exception:
75         r = rating
76     print(f"Rating: {rating} => {classify_feedback_match(r)}")
77
78 print("\nTesting dictionary mapping implementation:")
79 for rating in test_ratings:
80     try:
81         r = int(rating)
82     except Exception:
83         r = rating
84     print(f"Rating: {rating} => {classify_feedback_dict(r)}")
```

Terminal Output:

```
Testing dictionary mapping implementation:
Rating: 5 => Positive
Rating: 4 => Positive
Rating: 3 => Neutral
Rating: 2 => Negative
Rating: 1 => Negative
Rating: 0 => Invalid rating
Rating: -1 => Invalid rating
Rating: 3 => Neutral
```

OUTPUT :

This screenshot shows the terminal output from the previous code execution. It displays the results for the dictionary mapping implementation, showing that ratings 5 and 4 are 'Positive', 3 is 'Neutral', 2 and 1 are 'Negative', and 0 and -1 are 'Invalid rating'.

```
Testing dictionary mapping implementation:
Rating: 5 => Positive
Rating: 4 => Positive
Rating: 3 => Neutral
Rating: 2 => Negative
Rating: 1 => Negative
Rating: 0 => Invalid rating
Rating: -1 => Invalid rating
Rating: 3 => Neutral
```

Task Description #4 (Loops – Prime Numbers in a Range)

- Task: Generate a function using AI that displays all prime numbers within a user-specified range (e.g., 1 to 500).
- Instructions:
 - Get AI-generated code to list all primes using a for loop.
 - Analyze the correctness and efficiency of the prime-checking logic.
 - Ask AI to regenerate an optimized version (e.g., using the square root method).

Expected Output #4:

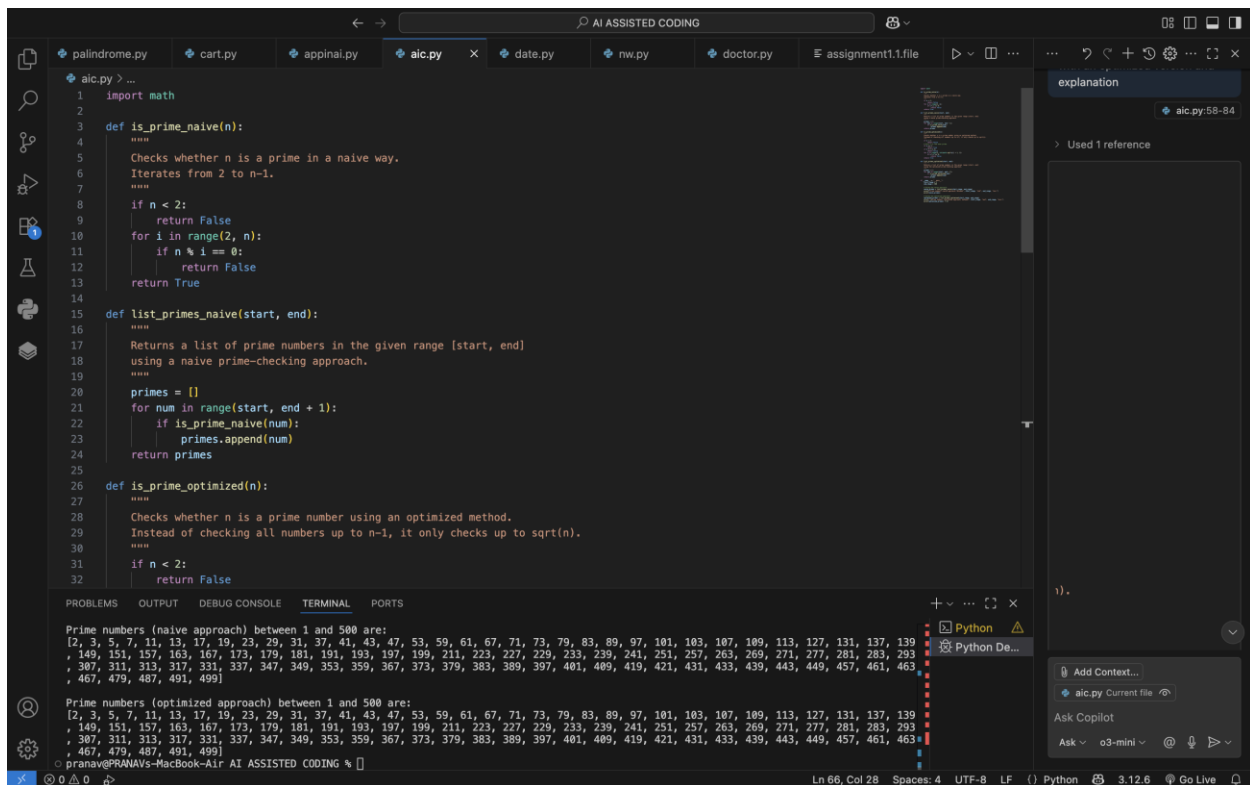
- Python program that lists all prime numbers within a given range, with an optimized version and explanation

PROMPT USED :

To Generate a function that displays all prime numbers within a user-specified range (e.g., 1 to 500)

To list all primes using a for loop.

CODE :



```
1 import math
2
3 def is_prime_naive(n):
4     """
5     Checks whether n is a prime in a naive way.
6     Iterates from 2 to n-1.
7     """
8     if n < 2:
9         return False
10    for i in range(2, n):
11        if n % i == 0:
12            return False
13    return True
14
15 def list_primes_naive(start, end):
16     """
17     Returns a list of prime numbers in the given range [start, end]
18     using a naive prime-checking approach.
19     """
20    primes = []
21    for num in range(start, end + 1):
22        if is_prime_naive(num):
23            primes.append(num)
24    return primes
25
26 def is_prime_optimized(n):
27     """
28     Checks whether n is a prime number using an optimized method.
29     Instead of checking all numbers up to n-1, it only checks up to sqrt(n).
30     """
31    if n < 2:
32        return False
```

Prime numbers (naive approach) between 1 and 500 are:

```
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]
```

Prime numbers (optimized approach) between 1 and 500 are:

```
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]
```



```
25
26 def is_prime_optimized(n):
27     """
28     Checks whether n is a prime number using an optimized method.
29     Instead of checking all numbers up to n-1, it only checks up to sqrt(n).
30     """
31     if n < 2:
32         return False
33     # Only 2 is the even prime.
34     if n == 2:
35         return True
36     if n % 2 == 0:
37         return False
38     for i in range(3, int(math.sqrt(n)) + 1, 2):
39         if n % i == 0:
40             return False
41     return True
42
43 def list_primes_optimized(start, end):
44     """
45     Returns a list of prime numbers in the given range [start, end]
46     using the optimized prime-checking approach.
47     """
48     primes = []
49     for num in range(start, end + 1):
50         if is_prime_optimized(num):
51             primes.append(num)
52     return primes
53
54 if __name__ == "__main__":
55     start_range = 1
56     end_range = 500
```

Prime numbers (naive approach) between 1 and 500 are:
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]

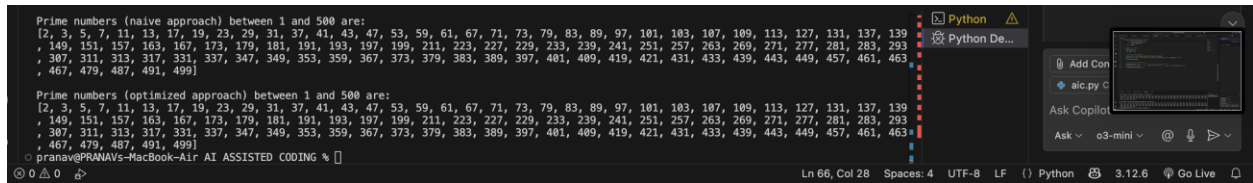
Prime numbers (optimized approach) between 1 and 500 are:
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]

```
43 def list_primes_optimized(start, end):
44     if is_prime_optimized(num):
45         primes.append(num)
46     return primes
47
48 if __name__ == "__main__":
49     start_range = 1
50     end_range = 500
51
52     # Using the naive approach:
53     naive_primes = list_primes_naive(start_range, end_range)
54     print("Prime numbers (naive approach) between", start_range, "and", end_range, "are:")
55     print(naive_primes)
56
57     # Using the optimized approach:
58     optimized_primes = list
59     print("\nPrime numbers < 1/1 > Accept [tab] Accept Word [tab] ... )
60     print(optimized_primes] 7.25
```

Prime numbers (naive approach) between 1 and 500 are:
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]

Prime numbers (optimized approach) between 1 and 500 are:
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499]

OUTPUT :

A screenshot of a code editor interface. The main window displays two lists of prime numbers between 1 and 500. The first list is labeled 'Prime numbers (naive approach) between 1 and 500 are:' and contains a long list of primes. The second list is labeled 'Prime numbers (optimized approach) between 1 and 500 are:' and contains a shorter list of primes. To the right of the code editor is a sidebar with a 'Python' tab and a 'Python De...' tab. Below these tabs is a search bar and a list of files, including 'a1c.py'. At the bottom of the sidebar is a section labeled 'Ask Copilot' with a search bar and a list of suggestions, including 'o3-mini'. The status bar at the bottom of the editor shows 'Ln 66, Col 28', 'Spaces: 4', 'UTF-8', 'LF', 'Python', '3.12.6', and 'Go Live'.

Task Description #5 (Classes – Library System)

- Task: Use AI to build a Library class with methods to `add_book()`, `issue_book()`, and `display_books()`.

- Instructions:

- o Generate Library class code using AI.

- o Analyze if methods handle edge cases (e.g., issuing unavailable books).

- o Ask AI to add comments and documentation.

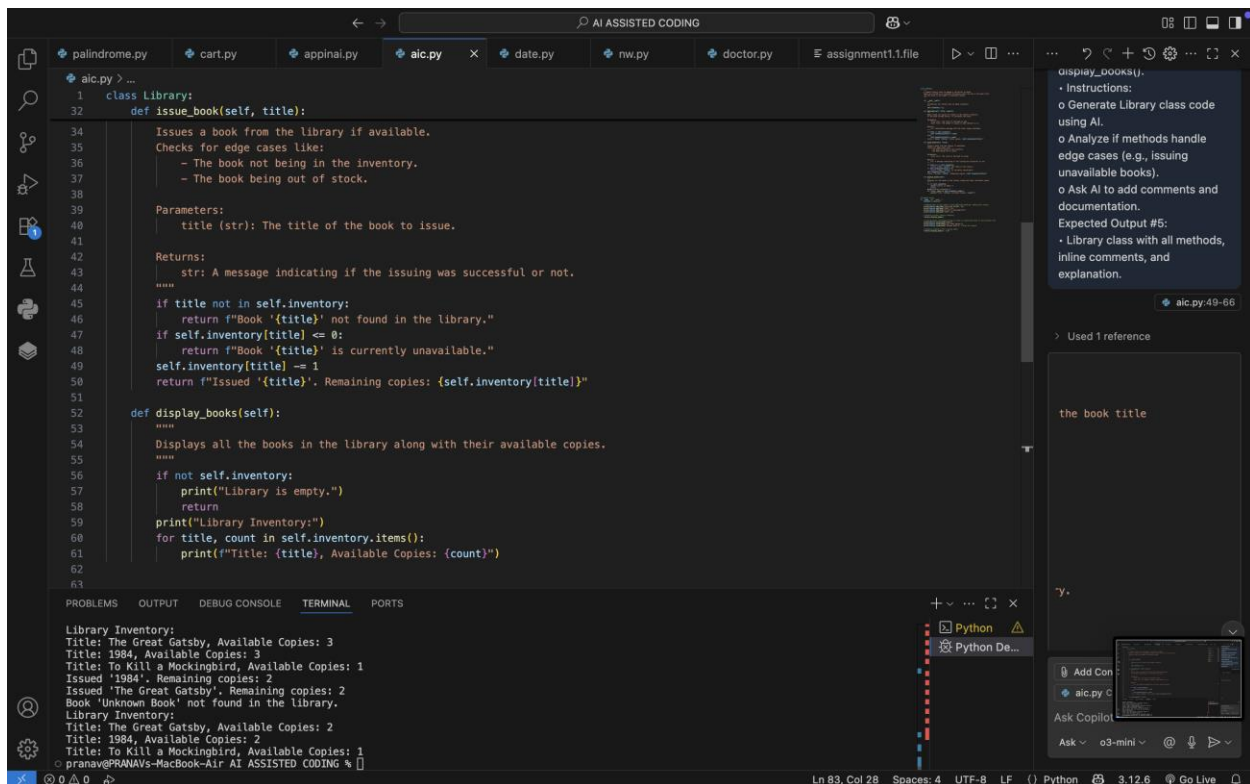
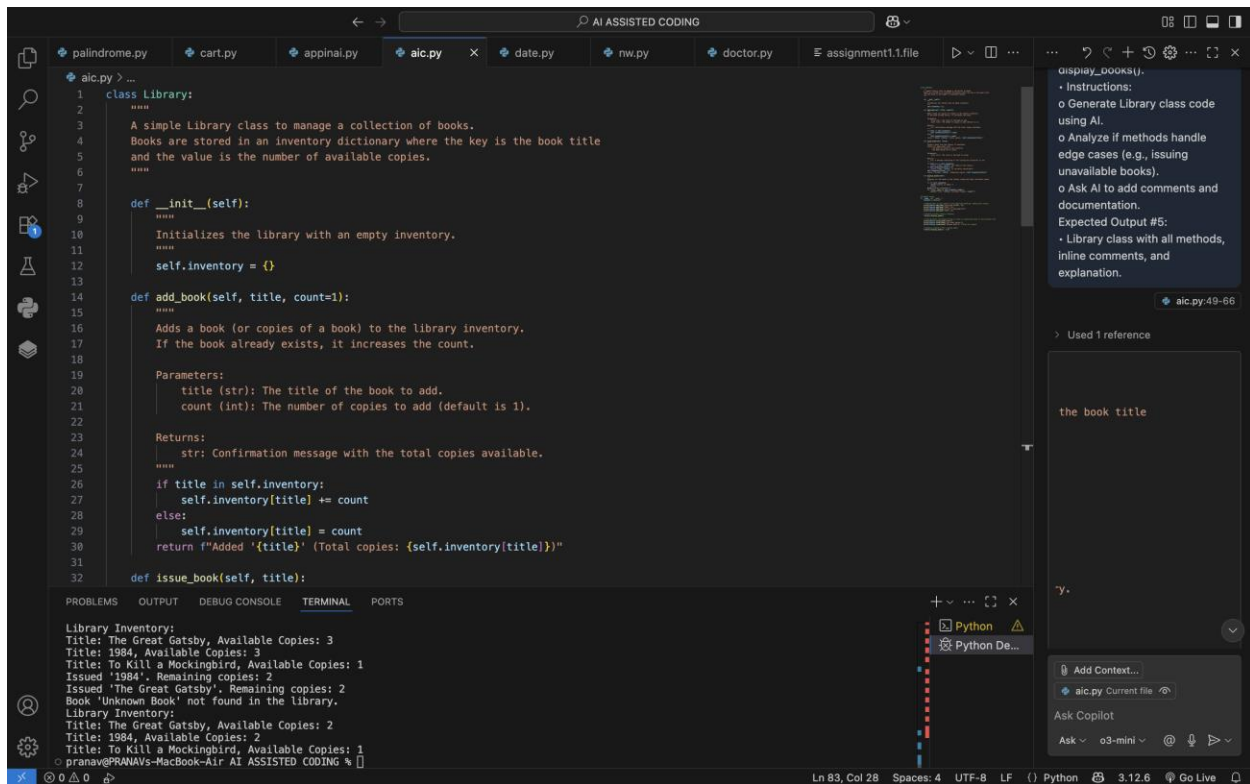
Expected Output #5:

- Library class with all methods, inline comments, and explanation.

PROMPT :

To build a Library class with methods to `add_book()`, `issue_book()`, and `display_books()`

CODE :



```
1 class Library:
2     def display_books(self):
3         """
4         """
5         if not self.inventory:
6             print("Library is empty.")
7             return
8         print("Library Inventory:")
9         for title, count in self.inventory.items():
10            print(f"Title: {title}, Available Copies: {count}")
11
12 # Example usage:
13 if __name__ == "__main__":
14     library = Library()
15
16     # Adding books to the library (with edge-case handling: adding more copies)
17     print(library.add_book("The Great Gatsby", 3))
18     print(library.add_book("1984", 2))
19     print(library.add_book("To Kill a Mockingbird"))
20     print(library.add_book("1984", 1))
21
22     # Display current library inventory
23     library.display_books()
24
25     # Issuing books (including trying to issue an unavailable book or non-existent one)
26     print(library.issue_book("1984"))
27     print(library.issue_book("The Great Gatsby"))
28     print(library.issue_book("Unknown Book")) # Book not present
29
30     # Display inventory after
31     library.display_books()
```

Library Inventory:
Title: The Great Gatsby, Available Copies: 3
Title: 1984, Available Copies: 3
Title: To Kill a Mockingbird, Available Copies: 1
Issued '1984'. Remaining copies: 2
Issued 'The Great Gatsby'. Remaining copies: 2
Book 'Unknown Book' not found in the library.
Library Inventory:
Title: The Great Gatsby, Available Copies: 2
Title: 1984, Available Copies: 2
Title: To Kill a Mockingbird, Available Copies: 1
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %

Instructions:
o Generate Library class code using AI.
o Analyze if methods handle edge cases (e.g., issuing unavailable books).
o Ask AI to add comments and documentation.
Expected Output #5:
• Library class with all methods, inline comments, and explanation.

Used 1 reference

the book title

OUTPUT :

```
Library Inventory:
Title: The Great Gatsby, Available Copies: 3
Title: 1984, Available Copies: 3
Title: To Kill a Mockingbird, Available Copies: 1
Issued '1984'. Remaining copies: 2
Issued 'The Great Gatsby'. Remaining copies: 2
Book 'Unknown Book' not found in the library.
Library Inventory:
Title: The Great Gatsby, Available Copies: 2
Title: 1984, Available Copies: 2
Title: To Kill a Mockingbird, Available Copies: 1
pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %
```